

T1. Freq. domain analysis and design



Q1.

$$\begin{aligned} \text{From } I\ddot{\theta} &= \sum T \\ &= \tau - \frac{1}{2}mg\sin\theta - c\dot{\theta} \end{aligned}$$

When $\theta \ll 1$, then $\sin\theta \approx \theta$, therefore:

$$\ddot{\theta} + \frac{c}{I}\dot{\theta} + \frac{mg}{2I}\theta = \frac{1}{I}\tau$$

Q2.

$$\Theta(s) \left(s^2 + \frac{c}{I}s + \frac{mg}{2I} \right) = \frac{1}{I}T(s)$$

$$\Rightarrow P(s) = \frac{T(s)}{\Theta(s)} = \frac{\frac{1}{I}}{s^2 + \frac{c}{I}s + \frac{mg}{2I}}$$

$$\text{Therefore, } \omega_n = \sqrt{\frac{mg}{2I}}, \quad \xi = c/(2I\omega_n)$$

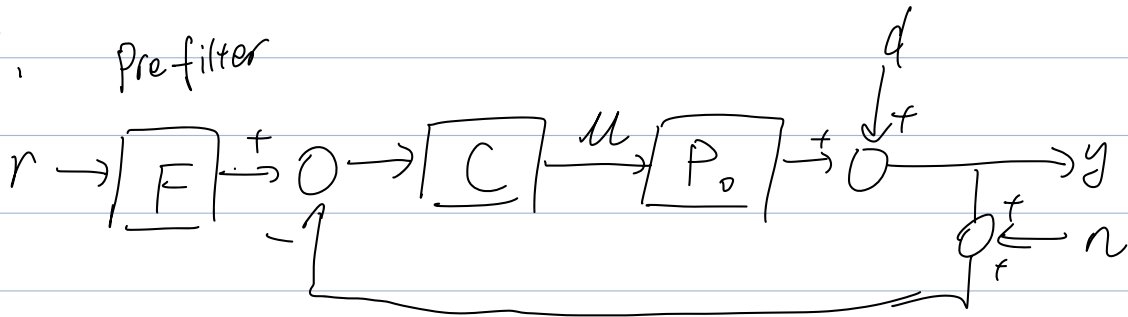
$$\text{i.e., } P(s) = \frac{1/I}{s^2 + 2\omega_n\xi s + \omega_n^2}$$

Q3.

$$\text{Q4. } \dot{x}(t) = \begin{bmatrix} \dot{\theta} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} -\frac{c}{I} & -\frac{mg}{2I} \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} \frac{1}{I} \\ 0 \end{bmatrix} \tau$$

$$y(e) = \theta(e) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \theta^* \\ \theta \end{bmatrix} + 0 \cdot e_0$$

Q1. Prefilter



$$(Fr - y) \cdot CP_0 = y \Rightarrow y = \frac{CP_0}{1 + CP_0} \cdot Fr.$$

r : reference input.

u : control signal.

y : output signal.

n : measurement noise.

Q2.

error can be made insensitive^{*} to disturbance^{*}
attenuated.

^{*} changes in plant
and operating

condition

unstable sys can be stabilized